

Qatar-2b

R-0694

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250425 · created 2026-07-27T20:56:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460790.8821 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.227 +/- 0.065
Transit depth (Rp/Rs)^2	5.15 +/- 2.97 [%]
Orbital Inclination (inc)	88.2 +/- 1.76
Airmass coefficient 1 (a1)	1.1 +/- 0.74
Airmass coefficient 2 (a2)	-0.07 +/- 0.68
Scatter in the residuals of the lightcurve fit is	74.04 %
Best Comparison Star	#3 - [49, 447]
Optimal Aperture	4.22
Optimal Annulus	13.35
Transit Duration (day)	0.0792 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.227 ± 0.065 vs 0.16327 (deviation 0.98σ)
Duration fit vs published	0.0792 d vs 0.0767 d (deviation 0.48σ)
Mid-time O–C	-4.7 min
Depth SNR	3.5
Residual scatter	74.04 %

Light curve

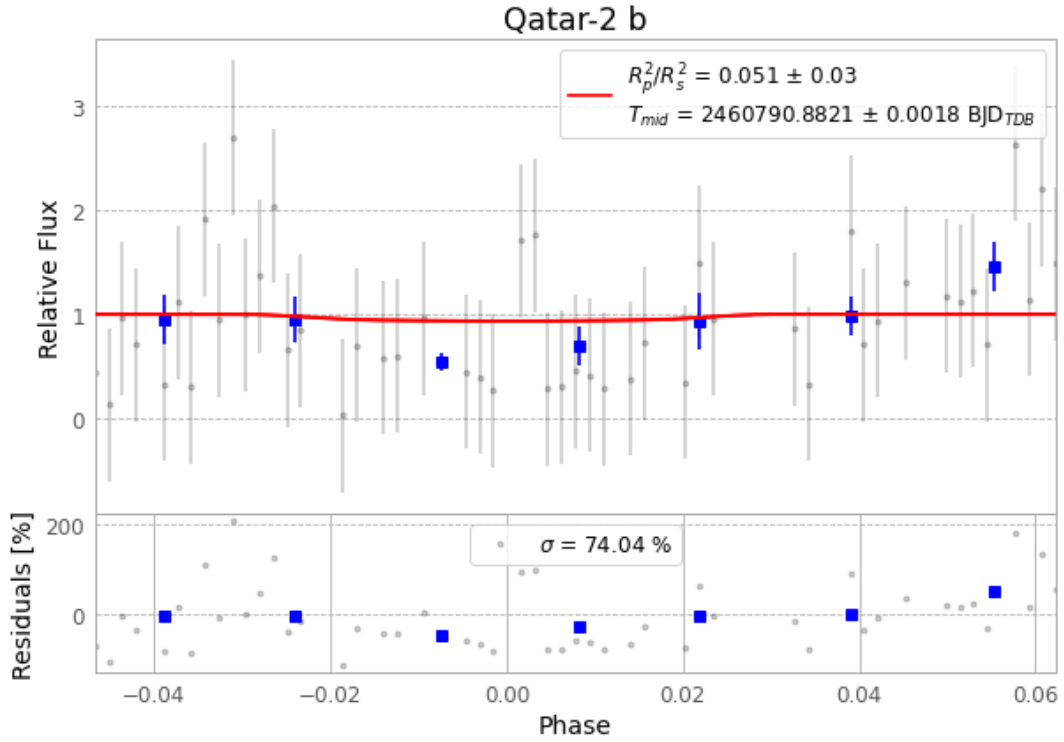


Figure 1 — FinalLightCurve_Qatar-2 b_2025-04-25.png (SHA-256 0a6cf871b527cd41...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 422], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a1120017086a8de4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

71 FITS frames (47 MB), Qatar-2250425073015.FITS → Qatar-2250425110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250425.log (b157eea34b89...), FinalLightCurve_Qatar-2 b_2025-04-25.png (0a6cf871b527...), FinalLightCurve_Qatar-2 b_2025-04-25.csv (bae4375a1dba...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:56Z.